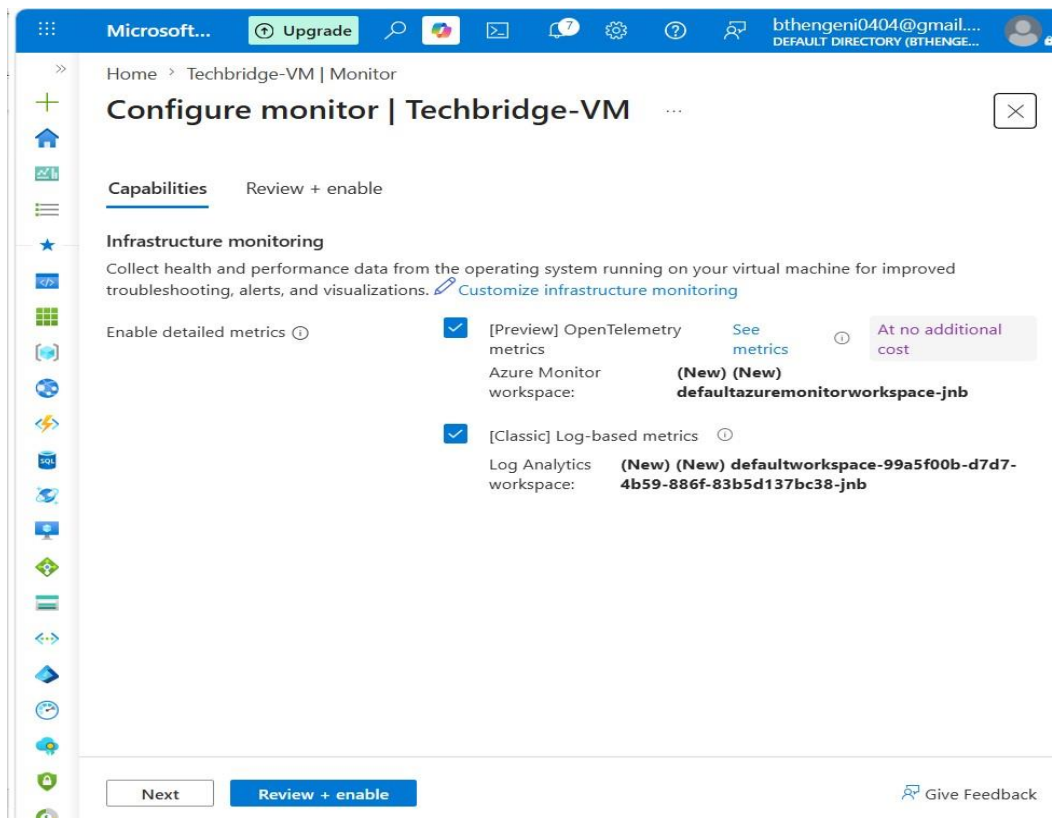


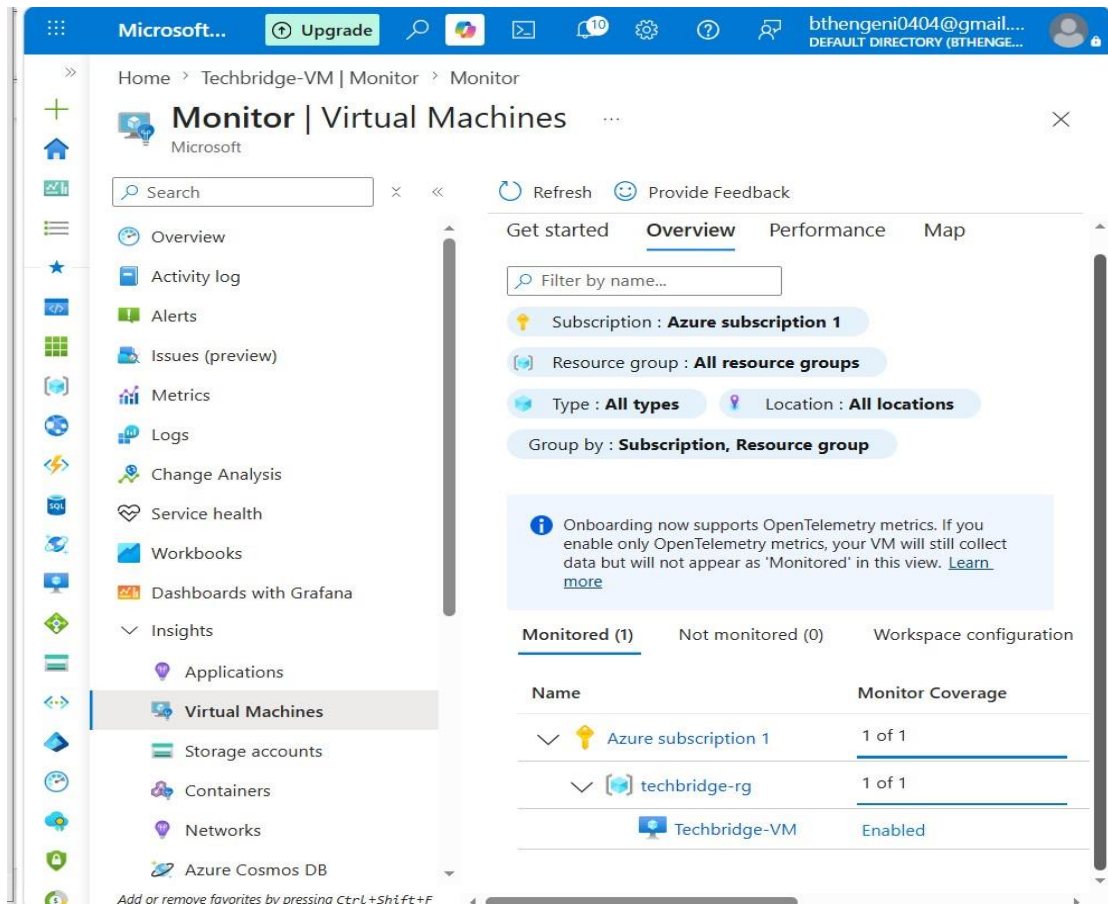
Task 1 — Enable Azure Monitor

Task 1.1 — Enable Monitoring on Your Virtual Machine

1. **Navigate:** Open the **Azure Portal**, go to **Virtual Machines**, and select your specific VM which is the **Techbridge-VM**
2. **Explore:** On the left-side menu, scroll down to the **Monitoring** section.
3. **Enable VM Insights:** Click on **Insights** and then select **Enable**. You will be prompted to choose a **Log Analytics Workspace**. If you don't have one, Azure will suggest creating a default one] *Log Analytics workspace:(New) (New) defaultworkspace-99a5f00b-d7d7-4b59-886f-83b5d137bc38-jnb*
4. **Configure Diagnostics:** Go to **Diagnostic settings**. Click **Enable guest-level monitoring**. Under the **Metrics** tab, ensure **CPU**, **Memory**, and **Disk I/O** are checked. Click **Save**.

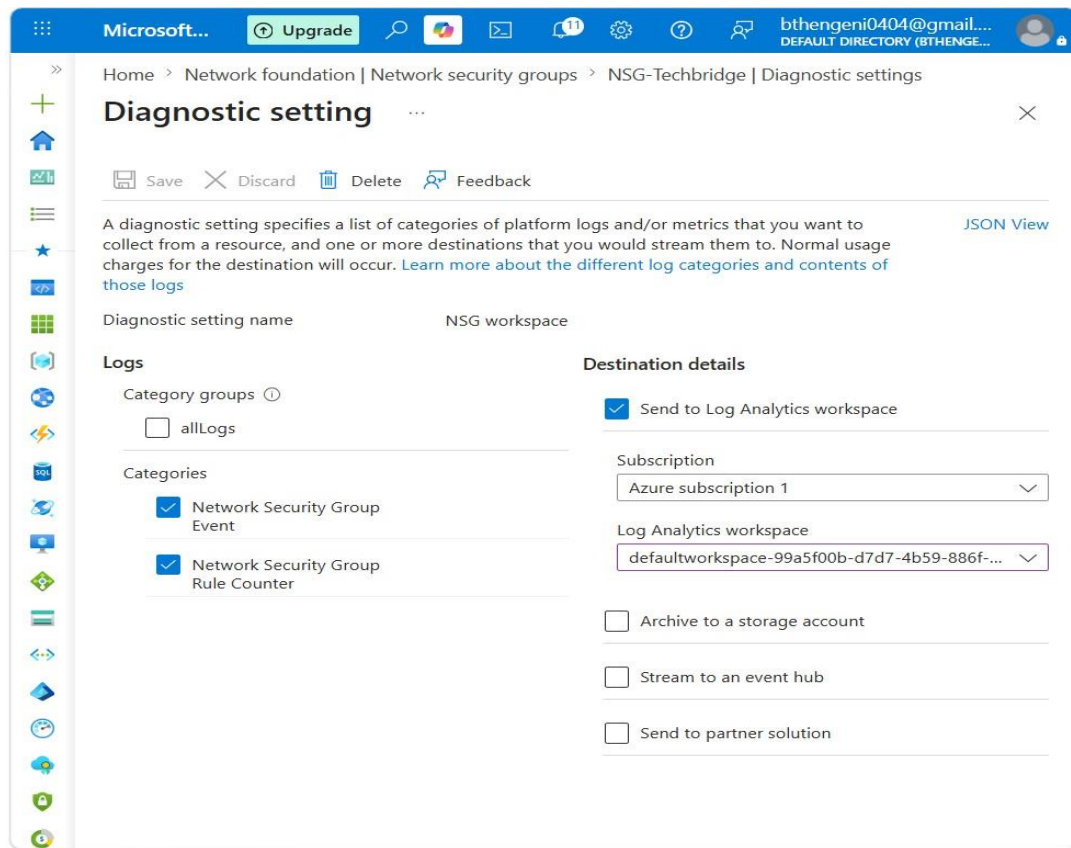
I enabled VM Insights and Diagnostic Settings for the virtual machine. The collected (performance metrics and guest-level logs) is stored in a **Log Analytics Workspace**, which acts as a centralized repository for querying and long-term data retention.





Task 1.2 — Enable Monitoring on Network Resources

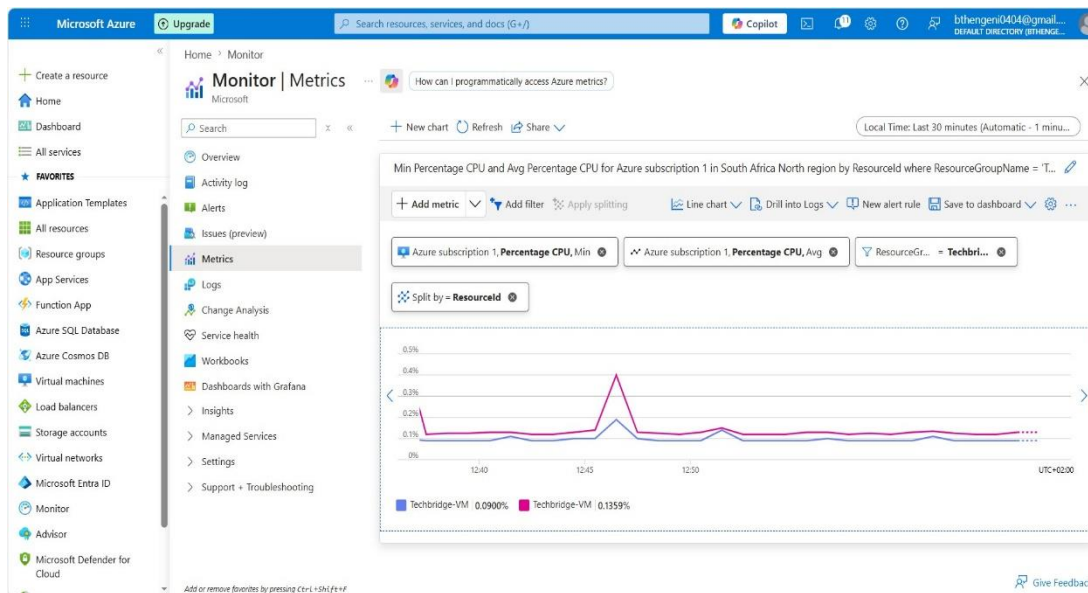
1. **NSG Logs:** Navigate to your **Network Security Group (NSG)**. Under **Monitoring**, select **Diagnostic settings > Add diagnostic setting**.
 - **Categories:** Select **NetworkSecurityGroupEvent** and **NetworkSecurityGroupRuleCounter**.
 - **Destination:** Check **Send to Log Analytics workspace** and select the workspace used in Task 1.1.
2. **VNet Metrics:** Navigate to your **Virtual Network**. Select **Diagnostic settings > Add diagnostic setting**. Select **AllMetrics** and send them to the same Log Analytics workspace.



Task 2 — Review Key Metrics

Task 2.1 — CPU Metrics

1. Search for **Monitor** in the top search bar and select **Metrics**.
2. **Scope:** Select your Subscription, Resource Group, and **Virtual Machine**.
3. **Metric:** Select **Percentage CPU**.
4. **Time Range:** Set it to **Last 30 minutes** or **Last 1 hour** to see the "spike" from your test load.

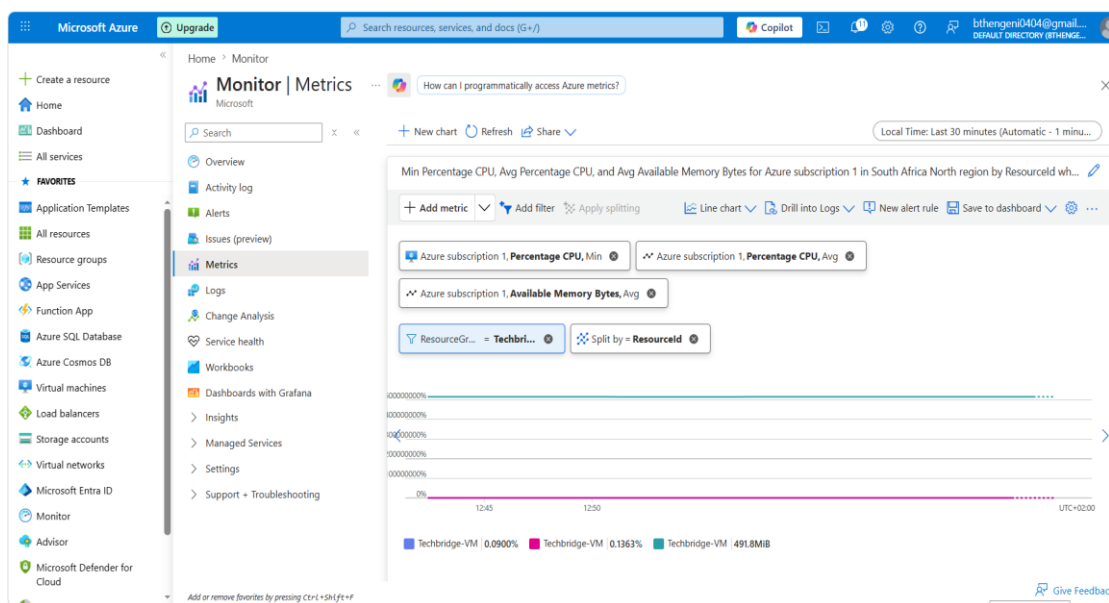


Observation:

The CPU metric shows a baseline of roughly 2–5% with a visible spike reaching 40% during the 'stress' test. This is expected as the VM was idle before the manual load was applied.

Task 2.2 — Network Metrics

1. In the same Metrics view, change the metric to **Network In Total** and **Network Out Total**.
2. **Formatting:** Click **Add metric** to show both on the same graph.
3. **Unit:** This is typically measured in **Bytes** (or MiB/GiB depending on scale).

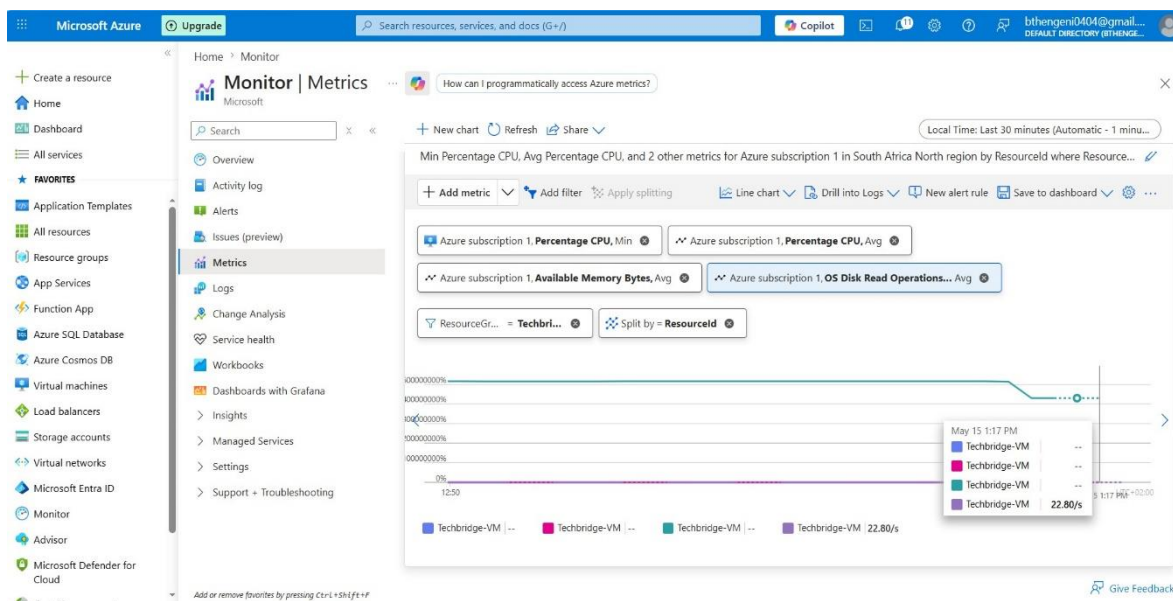


Observation:

Both the Inbound and Outbound traffic show activity. Inbound traffic spiked when I ran the update commands, while Outbound traffic remained lower, representing the acknowledgments sent back to the server.

Task 2.3 — Disk Metrics

1. Change the metric to **OS Disk Write Operations/Sec** or **Disk Write Bytes**.
2. Observe the correlation between disk activity and your previous tasks.



Observation:

Disk write activity peaked during the installation of packages. The metrics reveal that while read activity is constant for OS operations, write activity is highly bursty and tied to specific user actions.

Task 3 — Document Your Monitoring Setup

Task 3.1 — Monitoring Summary

Section	Details
Resources Monitored	Using VM Insights (Agents) for CPU/RAM/Disk. With NSG we use Diagnostic settings for security flow logs and Traffic metrics.
Data Retention	Data is stored in the Log Analytics Workspace . By default, retention which is often set to 30 days . For older data, I would need to increase the retention setting in the

	Workspace or using export logs to an Azure Storage Account .
Metric Observations	CPU and Disk metrics remained within healthy baselines. An "abnormal" baseline would be a sustained with 90%+ CPU usage without a known scheduled task, which could indicate a process hang or security breach

Reflection Questions

Question	Your Response
Importance of Cloud Monitoring	In the cloud we do not have any physical access to hardware. Monitoring provides the visibility needed to ensure availability, security, and performance across distributed resources allocated.
Metric vs. Log	A Metric is a numerical value over time such as the CPU used for real-time alerts. A Log is a record of a specific event ("User Login Failed") used for deep troubleshooting and auditing
High CPU at 2am Investigation	I would check the Process tab in VM Insights to see which specific application was running at anytime available. I would also check for scheduled tasks such as backups, antivirus scans, or "cron jobs" that would trigger at that time.
Cost-Optimisation	If monitoring shows a VM's CPU never exceeds 10%, I can "right-size" the resource by switching to a smaller, cheaper VM SKU to save money.